About the Project

Robust Potato Leaf Disease Identification Using Field-Collected Images

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Project Background

Deep learning approaches for plant disease detection often report high accuracy when trained and tested on curated datasets. However, real-world agricultural deployment remains challenging due to **uncontrolled environmental factors** such as variable lighting, complex backgrounds, partial occlusions, and heterogeneous disease symptoms.

This internship project focuses on **leveraging an existing, novel dataset of potato leaf diseases collected under uncontrolled field conditions**. The primary objective is to **analyze, benchmark, and deploy robust AI models** capable of performing reliably in real-world scenarios. Unlike many dataset-centric projects, **no new data collection is required**; students will work exclusively with the provided dataset and focus on **model performance, robustness, and deployment readiness**.

Project Objectives

1. **Conduct a focused literature review** on potato leaf disease detection and related plant disease datasets, with particular attention to datasets collected in uncontrolled or real-world environments, to position the provided dataset within existing research.
2. **Analyze and characterize the existing potato leaf disease dataset** captured in uncontrolled field conditions, including class distribution, visual variability, and environmental complexity.
3. **Develop and benchmark deep learning models** for potato leaf disease classification and/or localization using the provided dataset.
4. **Enhance model robustness** through data augmentation, architecture selection, and domain-aware training strategies suited for real-world agricultural imagery.
5. **Deploy the trained model** within a practical inference pipeline suitable for real-world use.Evaluate deployment performance, considering accuracy, inference latency, resource usage, and usability.

Technical Focus Areas

Dataset that will be used: https://data.mendeley.com/datasets/ptz377bwb8/1
Article about the data: https://www.sciencedirect.com/science/article/pii/S2352340923009861#refdata001

Students may focus on one or more of the following:

1. Comparative benchmarking of CNN vs Vision Transformer models
2. Robustness analysis under real-world noise and background variation
3. Lightweight or optimized models for edge or mobile deployment
4. Explainable AI (Grad-CAM, Score-CAM) for disease prediction transparency
5. End-to-end inference pipelines (image input → prediction → visualization)
6. Cross-dataset generalization analysis

Requirements

Mandatory

- Python programming
- Deep learning fundamentals
- Experience with PyTorch or TensorFlow

Preferred

- Computer vision and CNNs
- Model deployment (Flask, FastAPI, Streamlit, ONNX, or TensorRT)
- GPU-based training experience

Expected Outcomes

1. A fully trained and evaluated disease detection model
2. A deployable inference system (demo or prototype)Technical internship report
3. Journal type-report
4. Reusable deployment pipeline for future agricultural AI applications
5. Technical documentations

Timeline

Phase	Activities	Duration (approx.)
Phase 1	Literature review & dataset understanding	2 weeks
Phase 2	Model benchmarking & baseline experiments	4–6 weeks
Phase 3	Robustness improvement & optimization	4–6 weeks
Phase 4	Deployment pipeline development	4 weeks
Phase 5	Evaluation, documentation & research write-up	3–4 weeks